Sigmoid Activation Derivative:

Let sigmoid of z be, s(z)

$$= \frac{1}{1+e^{-x}} = (1+e^{-x})^{-1}$$

the derivative of sigmoid s'(z)

$$= -(1+e^{-x})^{-2} \times (-e^{-x})$$

$$= \frac{e^{-x}}{(1+e^{-x})^{2}}$$

$$= \frac{1}{1+e^{-x}} \times \frac{e^{-x}}{1+e^{-x}}$$

$$= s(z)\left(1 - \frac{1}{(1+e^{-x})}\right)$$

$$= s(z)(1-s(z))$$